LAB 03

Introduction to **Object-Oriented Programming**

Class and Object

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Lab03 Exercises

❖ Lab03.1. Quadratic equation

Write a program using Function to find roots of quadratic equation ax^2+ bx +c=0 when “a”, “b” and “c” are given by user input

Example01 - console:

```
Program for calculating roots of quadratic equation
ax^2+ bx +c=0
Input value of a: 2
Input value of b: 3
Input value of c: 1
X1 = -0.5, X2 = -1
```

Example02 - console:

```
Program for calculating roots of quadratic equation
ax^2+ bx +c=0
Input value of a: 2
Input value of b: 3
Input value of c: 4
Equation roots are complex!
```

❖ Lab03.2. N times of Input

Write a program to allow user to input *N times* of Number and find the values of average, max, min and sum:

Console:

```
How many number to be input?
Number of input: 5
Value #1: 3
Value #2: 55
Value #3: 26
Value #4: 95
Value #5: 43

Max: 95
Min: 3
Average: 44.4
Sum: 222
```

Lab03 Exercises

❖ Lab03.3. Nested Menu

Implement an application in Java to display and navigate Menu about “Phone Setting”

Console:

```
Phone Setting:
1. General                >
2. Wi-Fi                  >
3. Bluetooth              >
4. Mobile Data            >
5. Hotspot                >
6. Notification           >
0. Quit
```

3/4/5/6 Options

```
Title:
============================
The Feature is not available
============================
0. Back
```

1. General

```
General:
1. About                  >
2. Software update        >
3. Storage                >
0. Back
```

General > About

```
General > About:
Name                iPhone
Model               IXs
Version             18.5
0. Back
```

General > Software update

```
General > Software Update:
======================
Software is up to date
======================
0. Back
```

2. Wi-Fi

```
Wi-Fi:
Status                    On
Network                   I-Coffee
1. Other networks         >
0. Back
```

Wi-Fi > Other networks

```
Wi-Fi > Other networks:
Bayon coffee              *****
Angkor coffee             **
Brown coffee              ****
Koi                       *
0. Back
```

Lab03 Exercises

❖ Lab03.4. Infinite Input of Student

Write a program to allow user to *infinitely input student* to a table by using *Array of Student Class*.
You can suggest Attributes (id, name, age, etc.)

Console:

```
Student #1:
Id: 111
Name: Tola
Age: 17

Do you want to add more (y/n)?: y
Student #2:
Id: 222
Name: Makara
Age: 16

Do you want to add more (y/n)?: y
Student #2:
Id: 333
Name: Kompheak
Age: 25

Do you want to add more (y/n)?: n
```

Console:

```
Do you want to add more (y/n)?: n

=================================================
| No      | ID      | Name              | Age     |
=================================================
| 1       | 111     | Tola              | 17      |
| 2       | 222     | Makara            | 16      |
| 3       | 333     | Kompheak          | 25      |
=================================================
```

Lab03 Exercises

❖ Lab03.5. Static functions in Math Class I

Implement an application in Java to *create your Math class* with following return functions/methods:

- **add** (two parameters A and B)
- **subtract** (two parameters A and B)
- **multiply** (two parameters A and B)
- **divide** (two parameters A and B)
- **min** (two parameters A and B)
- **max** (two parameters A and B)

```
System.out.print(Math.add(5, 7));
System.out.print(Math.subtract(5, 7));
```

❖ Lab03.6. Static functions in Math Class II

Implement an application in Java to *create your Math class* with following return functions/methods:

- **factorial** (one parameter of a number)
- **rectangleSurface** (two parameters Width and Height)
- **circleSurface** (one parameter of radius)
- **max** (five parameters A, B, C, D, and E)
- **min** (five parameters A, B, C, D, and E)

Lab03 Exercises

❖ Lab03.7. Student Management

Write a program in Java to manage student by using Student Class and Array:

- **Attributes** (id, name, age, etc.)
- Must implement the following **methods** in Student class:
  - setValues()
  - display()

Console: (Option 1)

```
====== Menu ======
1. Create a student
2. List students
3. Quit
Choose an option: 1
Student #1:
Id: 111
Name: Tola
Age: 17
A student is added to the list

====== Menu ======
1. Create a student
2. List students
3. Quit
Choose an option:
```

Console: (Option 2)

```
====== Menu ======
1. Create a student
2. List students
3. Quit
Choose an option: 2

=================================================
| No      | ID      | Name              | Age     |
=================================================
| 1       | 111     | Tola              | 17      |
| 2       | 222     | Makara            | 16      |
| 3       | 333     | Kompheak          | 25      |
=================================================
```

Lab03 Exercises

❖ Lab03.8. Class diagrams to Java class

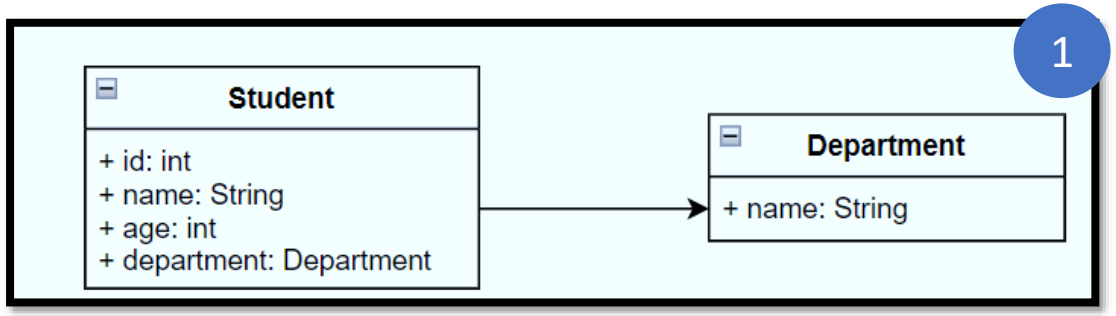
Create a model following class diagrams and objects:

- Phone shop: **Product** and **Category**
- Youtube: **User** and **Video**
- Book Store: **Book** and **Author**

1. Create a class diagram to represent relationship of two classes
2. Implement each **Java Class** with **at least 5 attributes** defined
3. Create **at least 1 objects** of **related Class**
4. Create **at least 3 objects** with values assigned
5. Create **an array** and put all objects to that array
6. Create **a loop to display** all object in that array

Lab03 Exercises

Follow the example in the figure to do your exercise 8:



2

```java
class Department {
    String name;
}

class Student {
    String id;
    String name;
    int age;
    Department dept;
    void setValues(String pId, String pName, int pAge, Department
        pDept) {
        id = pId;
        name = pName;
        age = pAge;
        dept = pDept;
    }
    void display() {
        System.out.println("ID: " + id + " Name: " + name + " Age: "
            + age + " Dept: " + dept.name);
    }
}
```

3

```java
public class SampleDemo {
    public static void main(String[] args) {
        // Create an array of Student
        Student[] stuArr = new Student[3];

        // Create related Class object
        Department dept = new Department();
        dept.name = "Computer Science";

        Department dept2 = new Department();
        dept2.name = "Economi";

        // Create an object of a student, set values and add to array
        Student stu1 = new Student();

        stu1.setValues("111", "Tola", 18, dept);
        stuArr[0] = stu1;

        Student stu2 = new Student();
        stu2.setValues("222", "Makara", 17, dept2);
        stuArr[1] = stu2;

        Student stu3 = new Student();
        stu3.setValues("333", "Kompheak", 23, dept);
        stuArr[2] = stu3;

        for(Student stu: stuArr) {
            stu.display();
        }
    }
}
```

Console:

```
ID: 111 Name: Tola Age: 18 Dept: Computer Science
ID: 222 Name: Makara Age: 17 Dept: Economi
ID: 333 Name: Kompheak Age: 23 Dept: Computer Science
```

Good luck 👌

✋ How to submit my Lab works???

👉 Screenshot your source codes and answers as a single **pdf** file and submit to the **Moodle** with Java file attachment as well.

Note: Don’t zip the files for submission